



Oncor Southern DFW Load Interconnection and General Grid Strengthening Project – ERCOT Independent Review Status Update

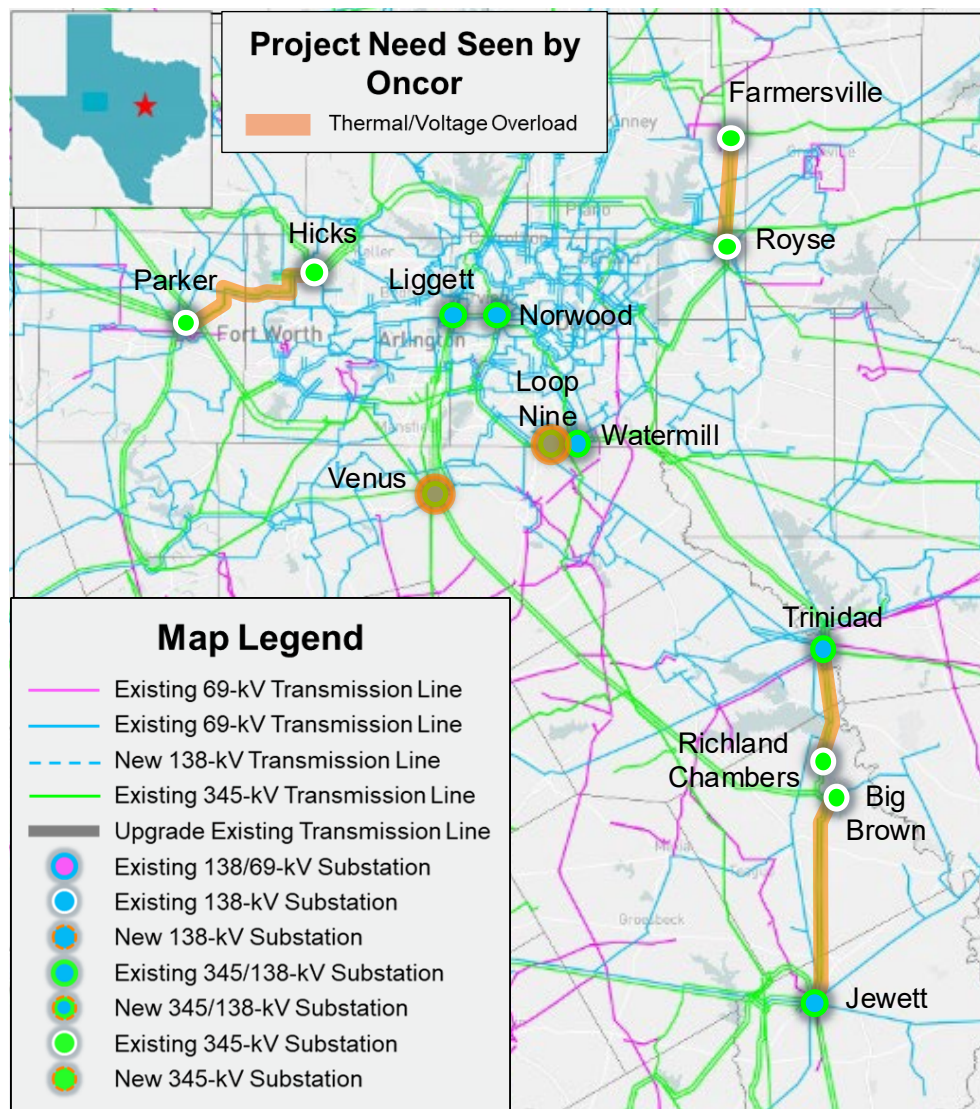
Tanzila Ahmed

RPG Meeting
July 29, 2025

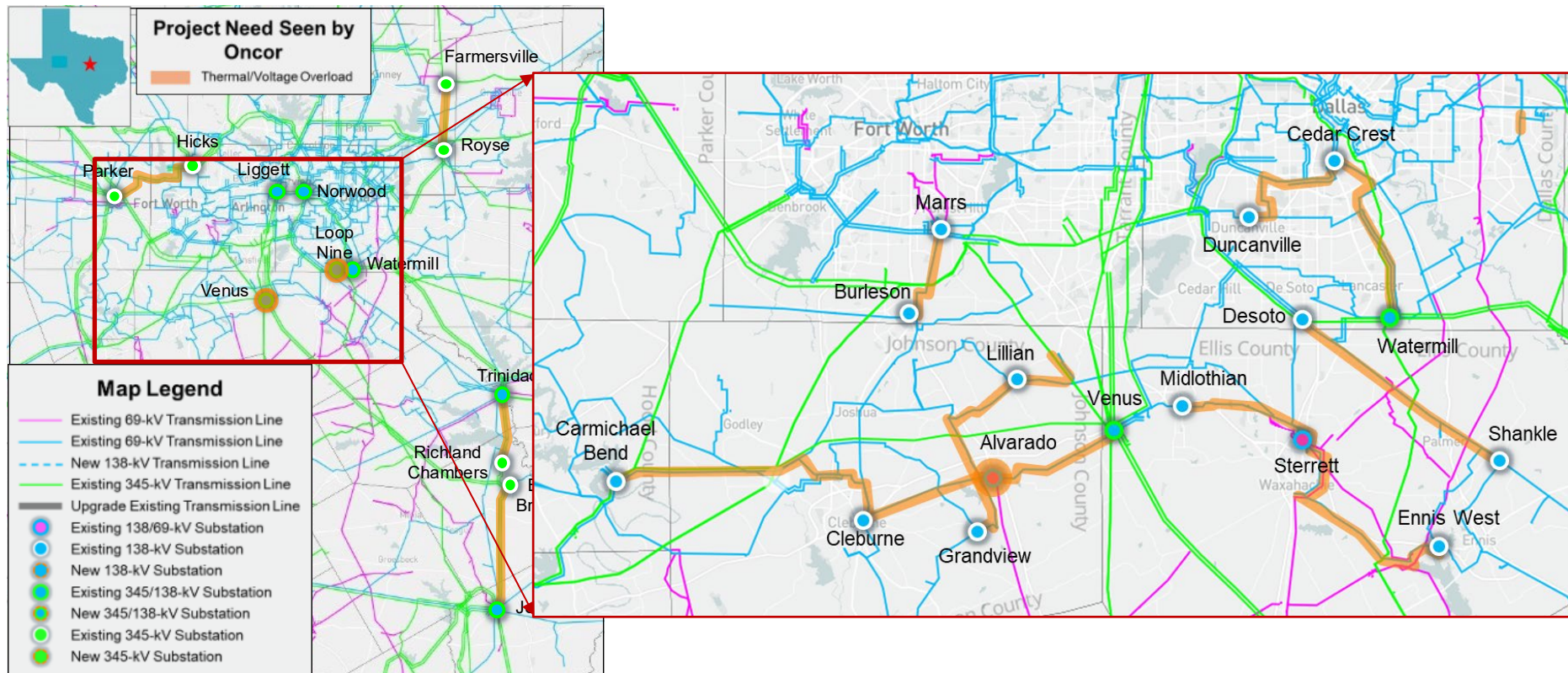
Recap – Introduction

- Oncor submitted the Southern Dallas Fort Worth (DFW) Load Interconnection and General Grid Strengthening Project for Regional Planning Group (RPG) review in February 2025
 - This Tier 1 project is estimated to cost \$1.219 billion and will require a Certificate of Convenience and Necessity (CCN) filing
 - Estimated in-service date (ISD) is December 2028
 - Addresses the thermal overloads and voltage violations due to proposed load additions in the Southern DFW in the North Central and East Weather Zones
- This project is currently under ERCOT Independent Review (EIR)
- Oncor presented project overview and ERCOT presented EIR Scope at the April RPG meeting
 - <https://www.ercot.com/calendar/04292025-RPG-Meeting>

Recap – Study Area Map with Violations Seen by Oncor on 345-kV System



Recap – Study Area Map with Violations Seen by Oncor on 138-kV System



Recap – Project Proposed by Oncor

- Construct a new Greene Road 345/138-kV Switch by installing fourteen 345-kV, 5,000 A and ten 138-kV, 3,200 A breakers in a breaker-and-a-half bus arrangement;
 - Re-terminate the existing Watermill Switch to Sargent Road Switch/West Levee Switch 345-kV double-circuit transmission line into the new Greene Road 345-kV Switch, which creates the new Watermill Switch to Greene Road Switch and Greene Road Switch to Sargent Road Switch/West Levee Switch 345-kV double-circuit transmission lines;
 - Re-terminate the existing Wilson Switch to Cedar Hill Switch/Cedar Crest Switch 138-kV double-circuit transmission line into the new Greene Road 138-kV Switch, which creates the new Wilson Switch to Greene Road Switch and Greene Road Switch to Cedar Hill Switch/Cedar Crest Switch 138-kV double-circuit transmission line;
 - Install two new 345/138-kV autotransformers with a normal rating of 700 MVA and an emergency rating of 750 MVA;
- Rebuild the new Greene Road Switch to Watermill Switch 345-kV double-circuit transmission line on double-circuit structures with both circuits in place, with normal and emergency ratings of at least 2,987 MVA, approximately 3.6-mile;
- Rebuild the new Wilson Switch to Greene Road Switch 138-kV double-circuit transmission line on double-circuit structures with both circuits in place, with normal and emergency ratings of at least 764 MVA, approximately 2.0-mile;
- Rebuild the new Greene Road Switch to Cedar Crest 138-kV single-circuit transmission line on double-circuit structures with one circuit in place, with normal and emergency ratings of at least 764 MVA, approximately 10.9-mile;

Recap – Project Proposed by Oncor (cont.)

- Construct a new Alba Road 345-kV Switch by installing eleven 345-kV, 5,000 breakers in a breaker-and-a-half bus arrangement;
 - Re-terminate the existing Watermill Switch to Greene Road Switch 345-kV double-circuit transmission line into the new Alba Road 345-kV Switch, which creates the new Watermill Switch to Alba Road Switch and Alba Road Switch to Greene Road Switch 345-kV double-circuit transmission lines;
- Construct a new Stainback 345-kV Switch by installing fourteen 345-kV, 5,000 breakers in a breaker-and-a-half bus arrangement;
 - Re-terminate the existing Watermill Switch to Elrod Switch/Big Onion Switch 345-kV double-circuit transmission line into the new Stainback 345-kV Switch, which creates the new Watermill Switch to Stainback Switch and Stainback Switch to Elrod Switch/Big Onion Switch 345-kV double-circuit transmission lines;
- Rebuild the new Watermill Switch to Stainback Switch 345-kV double-circuit transmission line on double-circuit structures with both circuits in place, with normal and emergency ratings of at least 2,987 MVA, approximately 3.0-mile;
- Rebuild the existing Watermill Switch to Wilson Switch 138-kV double-circuit transmission line on double-circuit structures with both circuit in place, with normal and emergency ratings of at least 764 MVA, approximately 0.8-mile;

Recap – Project Proposed by Oncor (cont.)

- Construct a new Ironwood 345/138-kV Switch by installing seventeen 345-kV, 5,000 A and ten 138-kV, 3,200 A breakers in a breaker-and-a-half bus arrangement;
 - Re-terminate the existing Liggett Switch to Endeavor Switch 345-kV Line at the new Ironwood 345-kV Switch, which creates the new Liggett Switch to Venus Switch (north bus)/Ironwood Switch 345-kV double-circuit transmission line;
 - Disconnect the existing Endeavor Switch to Venus Switch (south bus) and Midlothian ANP #2 to Venus Switch (south bus) 345-kV Lines from Venus Switch (south bus) and connect the Midlothian ANP to Endeavor 345-kV Switch. This will create Midlothian ANP #1 to Venus Switch (north bus) and Midlothian ANP #2 to Endeavor Switch 345-kV double-circuit transmission line;
 - Re-terminate the existing Timberview Switch to Venus Switch (south bus) at the new Ironwood 345-kV Switch, which creates the new Everman Switch to Venus Switch (north bus) and Timberview Switch to Ironwood Switch 345-kV double-circuit transmission line;
 - Re-terminate the existing Cedar Hill Switch to Venus Switch (south bus) at the new Ironwood 345-kV Switch, which creates the new Sherry Switch to Venus Switch (north bus) and Cedar Hill Switch to Ironwood Switch 345-kV double-circuit transmission line;
 - Re-terminate the existing Sam Switch to Venus Switch (south bus) at the new Ironwood 345-kV Switch, which creates the new Fort Smith Switch to Venus Switch (north bus) and Sam Switch to Ironwood Switch 345-kV double-circuit transmission line;
 - Re-terminate the existing Navarro Switch to Venus Switch (south bus) at the new Ironwood 345-kV Switch, which creates the new Navarro Switch to Venus Switch (north bus)/Ironwood Switch 345-kV double-circuit transmission line;
 - Re-terminate the existing Cottonwood Creek 345/138-kV Autotransformer #2 at the north bus of Venus 345-kV Switch by installing one 345-kV, 5,000 A breaker;
 - Install two 345/138-kV autotransformers with normal and emergency ratings of 700 MVA and 750 MVA, respectively;

Recap – Project Proposed by Oncor (cont.)

- Loop the existing Kemp Ranch Switch to Sardis Switch/Soap Creek 138-kV double-circuit transmission line into the new Ironwood 138-kV Switch by disconnecting the double-circuit line at structure #1/2 (Midlothian Tap) and constructing four circuits from Midlothian Tap to the new Ironwood 138-kV Switch on separate structures, with normal and emergency ratings of at least 764 MVA, approximately 2.0-mile each;
- Rebuild the two Kemp Ranch Switch to Midlothian Tap 138-kV single-circuit transmission line sections using two separate structures, with normal and emergency ratings of at least 764 MVA, approximately 0.5-miles each;
- Rebuild the existing Sterrett Switch to Midlothian TXI 138-kV single-circuit transmission line sections, with normal and emergency ratings of at least 764 MVA, approximately 11.8-mile;
- Rebuild the existing Ennis West Switch to Sterrett Switch 138-kV single-circuit transmission line, with normal and emergency ratings of at least 614 MVA, approximately 21.0-mile;

Recap – Project Proposed by Oncor (cont.)

- Rebuild the existing Big Brown 345-kV Switch by installing twelve 345-kV, 5,000 A breakers in a breaker-and-a-half bus arrangement. Upon completion, Big Brown Switch will be renamed as Pin Oak Switch;
 - Re-terminate the existing Big Brown Switch to Jewett Switch 345-kV double-circuit transmission line at the new Pin Oak 345-kV Switch, which creates the new Pin Oak Switch to Jewett Switch 345-kV double-circuit transmission line;
 - Re-terminate the existing Big Brown Switch to Navarro Switch 345-kV double-circuit transmission line at the new Pin Oak 345-kV Switch, which creates the new Pin Oak Switch – Navarro Switch 345-kV double-circuit transmission line;
 - Re-terminate the existing Big Brown Switch to Richland Chamber Switch 345-kV double-circuit transmission line at the new Pin Oak 345-kV Switch, which creates the new Pin Oak Switch to Richland Chambers Switch 345-kV double-circuit transmission line;
- Rebuild the new Jewett Switch to Pin Oak Switch 345-kV double-circuit transmission line on double-circuit structures with both circuits in place, with normal and emergency ratings of at least 1,792 MVA and with a conductor rating of at least 2,987 MVA, approximately 32.8-mile;
- Rebuild the existing Richland Chambers Switch to Trinidad Switch 345-kV double-circuit transmission line on double-circuit structures with both circuits in place with normal and emergency ratings of at least 1,792 MVA and a with conductor rating of at least 2,987 MVA, approximately 18.7-mile;

Recap – Project Proposed by Oncor (cont.)

- Rebuild the existing Parker Switch to Hicks Switch 345-kV transmission line, with normal and emergency ratings of at least 1,912 MVA and with a conductor rating of at least 2,987 MVA, approximately 23.0-mile;
- Rebuild the existing Pebble Creek Switch to Shankle Switch 138-kV transmission line, with normal and emergency ratings of at least 764 MVA, approximately 15.5-mile;
- Rebuild the existing Mesquite East Switch to Seagoville Switch 138-kV transmission line, with normal and emergency ratings of at least 764 MVA, approximately 7.4-mile;
- Rebuild the existing Farmersville Switch to Royse Switch 345-kV double-circuit transmission line on double-circuit structures with both circuits in place, with normal and emergency ratings of at least 1,792 MVA and with a conductor rating of at least 2,987 MVA, approximately 15.25-mile;
- Install one +250/-250 MVar STATCOM at each of the switch stations below. The STATCOMs are intended to be Grid-forming;
 - Greene Road 345-kV Switch
 - Alba Road 345-kV Switch
 - Wilmer 345kV Switch

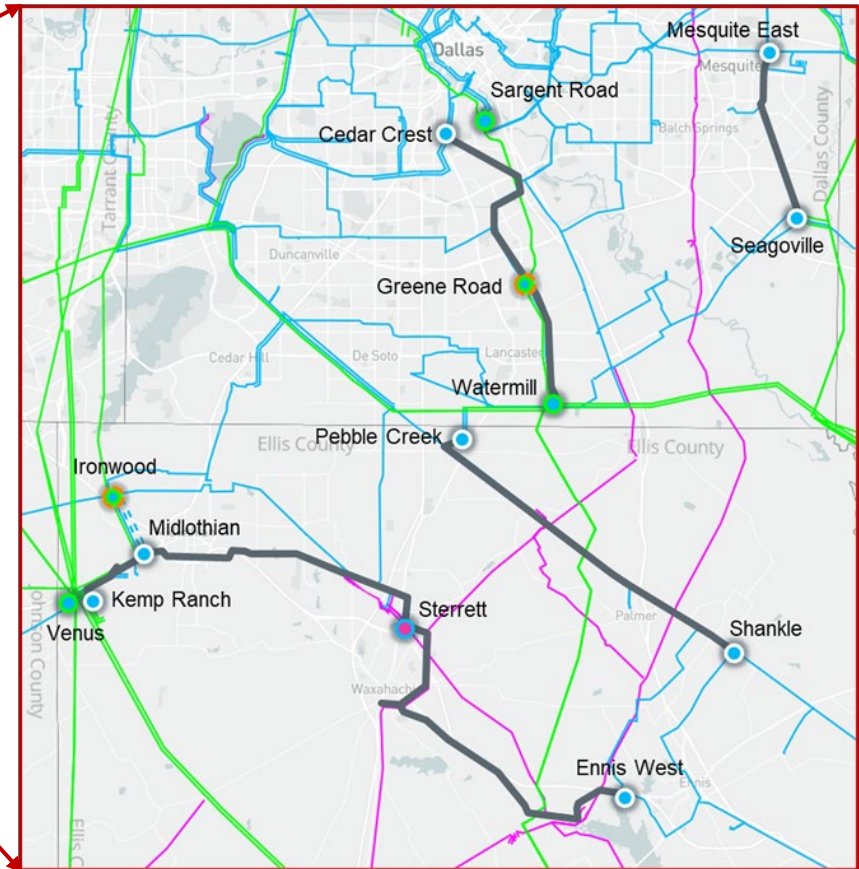
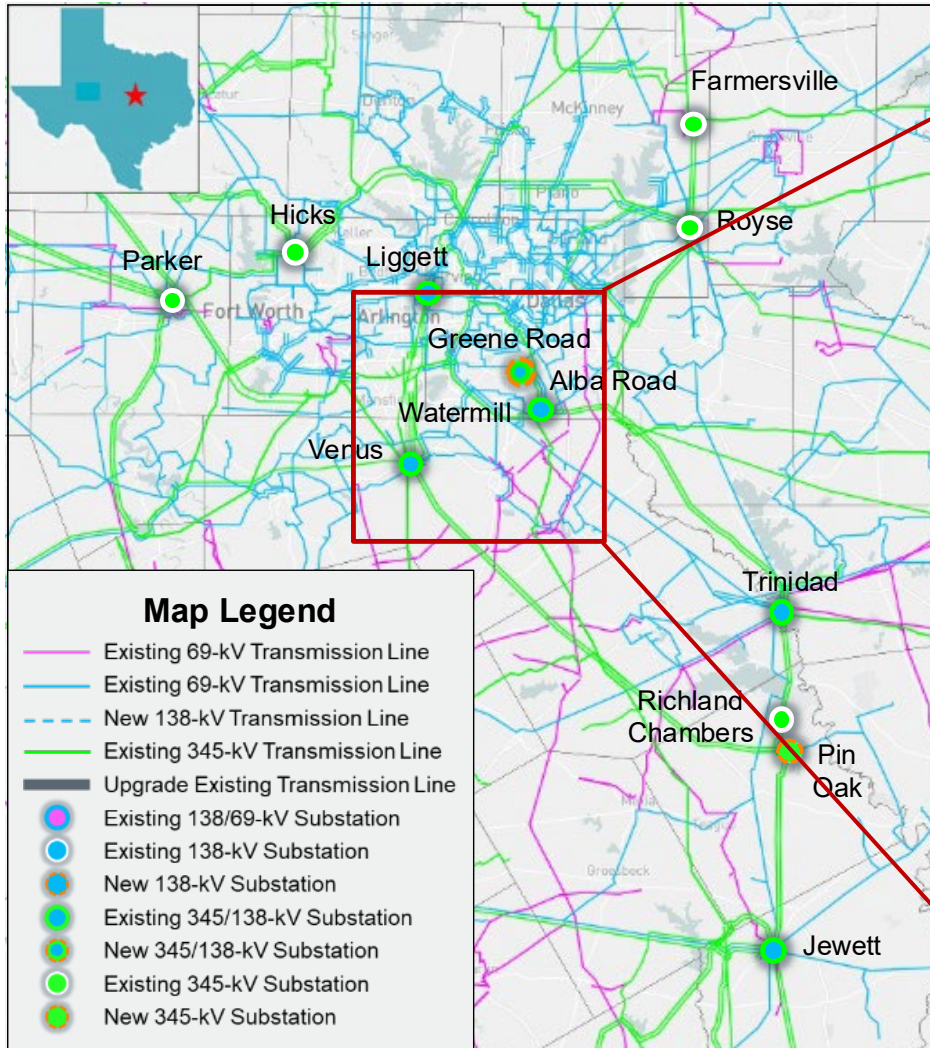
Recap – Project Proposed by Oncor (cont.)

- Install two 240 MVar 345-kV capacitor banks (3-80 MVar each) at;
 - Greene Road 345-kV Switch
 - Alba Road 345-kV Switch
 - Stainback 345-kV Switch
- Install two 110.4 MVar 138-kV capacitor banks (3-36.8 MVar each) at;
 - Greene Road 138-kVSwitch
 - Ironwood 138-kVSwitch
 - Pebble Creek 138-kVSwitch
- For existing 345-kV terminal equipment, ensure they meet or exceed a rating of at least 3,000 A;
- For the new 345-kV terminal equipment, ensure they meet or exceed a rating of 5,000 A if the station is 5,000 A capable. Otherwise ensure the new 345-kV terminal equipment meets or exceeds a rating of 3,200 A; and
- For the 138-kV terminal equipment, ensure they meet or exceed a rating of 3,000 A.

Recap – Map of Proposed Option with 345-kV Upgrades



Recap – Map of Proposed Option with 138-kV Upgrades



Recap – Study Assumptions

- Final 2024 Regional Transmission Planning (RTP) 2029 summer peak load case was used as the start case
- Transmission updates
 - New transmission projects, listed in the [Appendix A1](#), based on the February 2025 Transmission Project and Information Tracking (TPIT) report along with recently approved RPG projects were added to the study base case
 - Transmission projects, listed in the [Appendix A2](#), identified as placeholder projects in the 2024 RTP in the study area that have not been approved by RPG were removed from the study base case
 - Base case updates based on feedback from Oncor were also applied to the study base case

Recap – Study Assumptions (cont.)

- Generation updates

- New 6.9(1) generation, listed in [Appendix B](#), were added to the study base case based on the March 2025 Generator Interconnection Status (GIS) report
- All renewable generation were dispatched consistent with the 2024 RTP methodology
- GenHubs in North and North Central (NNC) Weather Zones were opened (turned off) to balance power

- Load updates

- All 2024 RTP Officer Letter Loads (OLLs) in the study area were removed from the study base case
- Large Loads including OLLs relevant to the project, provided by Oncor, in the study area were added to the study base case

- Reserve

- Load outside study Weather Zone was scaled to maintain the reserve consistent with the 2024 RTP

Status Update

- Reliability Need Analysis
 - N-1
 - G-1+N-1
 - G-1: Comanche Peak, Forney CCT, Martain Lake, Mt. Creek SES, and Midlothian generations
 - X-1+N-1
 - X-1: Desoto Switch (SW), Greene Road, Monticello SES, Sargent Road, Seagoville, Trinidad SES, Venus Kemp Ranch, Watermill, West Levee SW, and Wilmer 345/138-kV transformers
- Options Evaluation
 - N-1, G-1+N-1, and X-1+N-1

Preliminary Results of Reliability Assessment – Need Analysis

- ERCOT conducted steady-state load flow analysis for the study base case according to the NERC TPL-001-5.1 and ERCOT Planning Criteria to identify the project need

Contingency Category	Voltage Violations	Thermal Overloads	Unsolved Power Flow
N-0 (P0)	None	None	None
N-1 (P1, P2-1, P7)	19	31	1
G-1+N-1 (P3)*	1	1	None
X-1+N-1 (P6-2)**	42	24	3
Total	62	66	4

* G-1: Comanche Peak, Forney CCT, Martain Lake, Mt. Creek SES, and Midlothian generations

** X-1: Desoto Switch (SW), Monticello SES, Sargent Road, Seagoville, Trinidad SES, Venus Kemp Ranch, Watermill, West Levee SW, and Wilmer 345/138-kV transformers

Next Steps and Tentative Timeline

- ERCOT will continue with project evaluation and perform the following
 - Develop and test alternatives
 - Conduct Planned Maintenance Outage Evaluation
 - Conduct Long-Term Load-Serving Capability Assessment
 - Request TSPs to conduct Cost Estimate and Feasibility Assessment
- Additional analyses to be performed on the preferred option
 - Dynamic Analysis may be conducted
 - Congestion Analysis to ensure that the identified transmission upgrades do not result in new congestion within the study area
 - Generation Addition and Load Scaling Sensitivity Analyses
 - Planning Guide Section 3.1.3(4)
 - Subsynchronous Resonance (SSR) Assessment
 - Nodal Protocol Section 3.22.1.3(2)
- Tentative Timelines
 - Status updates at the future RPG meetings
 - Final recommendation – Q4 2025

Thank you!



Stakeholder comments also welcomed through:

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Appendix A1 – Transmission Projects Added

- List of recently approved RPG transmission projects added to study base case

RPG/TPIT No	Project Name	Tier	Project ISD	From County
24RPG016	Rand Area Loop Project	Tier 2	4/1/2027	Kaufman
24RPG017	Venus Switch to Sam Switch 345-kV Line Project	Tier 1	5/1/2026	Ellis, Hill
24RPG022	Wilmer 345/138-kV Switch Project	Tier 1	5/1/2026	Dallas
24RPG025	Gunter 345/138-kV Switch Project	Tier 3	12/1/2025	Ellis, Hill

Appendix A2 – Transmission Projects Removed

- List of placeholder transmission projects that are directly related to proposed project were removed from the study base case

Project ID	Project Name	County(s)
2024-E3	Big Brown SES West (3381) to Jewett (3391) 345-kV Line Upgrade and Substation Rebuilds	Henderson, Freestone
2024-E12	Trinidad SES (3124) to Richland Chambers (3134) 345-kV Line Upgrade	Freestone, Leon
2024-E18	Big Brown SES West (3381) to Jewett (3391) 345-kV Line Upgrade	Freestone, Leon
2024-NC13	Pebble Creek (2229) to Trumbull (221) to Gamma (12344) to Shankle Switch (12329) 138-kV Line Upgrades	Ellis
2024-NC14	Green Road (3069) to Ten Mile (2126) to Watermill (2429) to Reindeer (3065) 138-kV Line Upgrades	Dallas
2024-NC18	Ennis West Switch (2241) to Templeton (12320) to Waxahachie (2321) 138-kV Line Upgrades	Ellis, Waxahachie
2024-NC31	Royse Area 345-kV Line Upgrades and Substation Rebuilds	Collin, Hopkins, Rockwall, Fannin, Lamar
2024-NC63	Watermill Area 345-kV Line Additions and Reactive Support	Dallas, Navarro, Hill, Leon
2024-NC68	Batchler Road (2217) to Watermill (2427) 345-kV Line Upgrades	Dallas
2024-NC70	Miller Road (2632) 345/138-kV Substation Addition and 345-kV Lines Re-Termination	Dallas, Ellis
2024-NC85	Miller Road (2635) to Kemp Ranch Switch (2303) 138-kV double-circuit transmission line Upgrade	Ellis
2024-NC86	Greene Road (3063/3069) New 138-kV and 345-kV Line Additions and Substation Rebuilds	Dallas

Appendix A2 – Transmission Projects Removed (cont.)

- List of additional placeholder transmission projects in the study area that have not been approved by RPG were removed

Project ID	Project Name	County(s)
2024-E2	Shamburger (3103) to Elkton (3105) 345-kV Double Circuit Line Addition	Smith
2024-N11	Alla Hubbard (1757) to Epco POI (12468) 138-kV Line Upgrade	Collin, Grayson
2024-NC01	Glen Heights (217) to Sterett Road (2237) 138-kV Line Upgrade	Ellis
2024-NC04	Railport (442) Area 138-kV Rebuild	Ellis, Johnson
2024-NC06	Miller (824) to Newman (849) 138-kV Line Upgrade	Dallas
2024-NC07	Walnut (823) to Naaman (825) 138-kV Line Upgrade	Dallas
2024-NC08	Centerville (810) to Miller (824) 138-kV Line Upgrade	Dallas
2024-NC09	Walnut (823) to Newman (849) 138-kV Line Upgrade	Dallas
2024-NC10	Glen Heights (217) to Desoto Switch 8 (2424) 138-kV Line Upgrade	Ellis, Dallas
2024-NC11	Watermill (2427) to Tricorner (2432) to Trinidad (3123/3124) 345-kV Line Upgrades	Dallas, Henderson
2024-NC15	Waxahachie (2321) to Waxahachie North (2320) 138-kV Line Upgrade and Waxahachie (2335) to Waxahachie OCF (2343) 69-kV Line Upgrades	Ellis
2024-NC19	Climax TNP (37280) to Bridge Point RC (37245) 69-kV Line Upgrade	Collin
2024-NC21	Carrollton Northwest (2363) to South TNP (37100) to TI TNP (37080) 138-kV Line Upgrades	Dallas, Denton
2024-NC22	Waxahachie Pump 2 (2315) to Waxahachie North (2320) 138-kV Line Upgrade	Ellis

Appendix A2 – Transmission Projects Removed (cont.)

- List of additional placeholder transmission projects in the study area that have not been approved by RPG were removed

Project ID	Project Name	County(s)
2024-NC25	LTV Sub (2259) to Grand Prairie (2262) 138-kV Line Upgrade	Dallas
2024-NC27	Navarro (3478) to Haney (213) to Payne Battlecreek POI (888876) to Hubbard (3515) 138-kV Line Upgrades	Navarro, Hill
2024-NC28	Olympus Switch (3723) to Montfort Switch (3454) to Chatfield (200) 138-kV Line Upgrade	Henderson, Navarro
2024-NC29	Allen Switch (2514) to Pineforest POI (888854) 345-kV Line Upgrade	Collin, Hopkins
2024-NC34	Trinidad SES (3127) to Nipak Tap (3260) to Mankin Switch (3265) 138-kV Line Upgrade	Henderson, Navarro
2024-NC35	Firewheel (821) to Wylie Switch (833) 138-kV Line Upgrade	Dallas, Collin
2024-NC38	Seagoville Area Upgrades	Dallas
2024-NC41	Ben Davis (968) to GOBLN_8 (2497) 138-kV Line Upgrade	Dallas
2024-NC44	Cedar Creek 138-kV Area Line Upgrades	Henderson
2024-NC51	Waxahachie-Sterrett Area Upgrades	Ellis
2024-NC59	Lavon Switch 345/138-kV Switch Substation Expansion and Lavon Switch (2475) to Allen Switch (2511) 138-kV Line Addition	Collin, Rockwall
2024-NC64	Calmont Switch (1955) to Benbrook Switch (1874) 138-kV Line Upgrade	Tarrant
2024-NC66	Tri Corner (2432) to Forney Switch (2437) 345-kV Line Upgrade	Dallas, Kaufman

Appendix A2 – Transmission Projects Removed (cont.)

- List of additional placeholder transmission projects in the study area that have not been approved by RPG were removed

Project ID	Project Name	County(s)
2024-NC67	Elkton (3105) to Tri Corner (2432) 345-kV Line Additions and Upgrades	Dallas, Kaufman, Smith, Van Zandt
2024-NC71	Mitchell Bend Switch (1895) to Carmichael Bend Switch (2285) to Benbrook Switch (1873) 345-kV Line Upgrades	Tarrant, Hood, Grayson
2024-NC74	Midlothian TXI (2307) to Waxahachie Northwest (2309) 138-kV Line Upgrades	Ellis
2024-NC75	Primrose Fort Worth (2196) to Rocky Creek (1881) to Primrose Fort Worth (2197) 138-kV Line Upgrades	Tarrant
2024-NC76	Gunter 345/138-kV Switch Addition and Gunter (2236) to Collin (2370) 138-kV Line Addition	Cooke, Denton, Collin, Grayson
2024-NC77	Skyview 345/138-kV Switch Addition and Area Upgrade	Dallas, Tarrant
2024-NC79	Upgrade Benbrook Switch 345/138-kV (1869/18741) and (1870/18751) Transformers	Tarrant
2024-NC80	Desoto Switch 345/138-kV Transformer Addition (2431/2424) and Desoto Switch (2424) to Loop9 (2848) 138-kV Line Addition	Dallas
2024-NC82	Rocky Creek 345/138-kV Transformer Addition (1880/1881)	Tarrant
2024-NC83	Hempill (2164) to Mistletoe Heights (2173) 138-kV Line Upgrade	Tarrant
2024-NC84	Wedgewood Switch (2184) to Bryant Irvin (2182) 138-kV Line Upgrade	Tarrant

Appendix B – New Generation Projects Added

- List of new generation, that met the Planning Guide 6.9(1) condition, added to the study base case

GINR	Project Name	Fuel	Projected COD	Capacity (~MW)	County
21INR0359	Hickerson Solar	SOL	3/1/2026	316.3	Bosque
22INR0437	TORMES SOLAR	SOL	3/31/2027	382.1	Navarro
24INR0126	High Noon Storage	OTH	12/1/2027	94	Hill
24INR0188	Tehuacana Creek Solar SLF	SOL	3/10/2027	505.5	Navarro
24INR0189	Tehuacana Creek BESS SLF	OTH	3/10/2027	419	Navarro
24INR0355	Anatole Renewable Energy Storage	OTH	1/11/2026	207.8	Henderson
24INR0364	Pitts Dudik II	SOL	1/29/2026	30.2	Hill
24INR0498	Fort Watt Storage	OTH	4/20/2027	205.4	Tarrant
24INR0631	Radian Storage SLF	OTH	4/22/2025	160.3	Brown
25INR0018	Yellow Cat Wind	WIN	9/30/2026	301.2	Navarro
25INR0046	Blue Skies BESS	OTH	12/31/2027	306.3	Hill
25INR0231	Apache Hill BESS	OTH	11/15/2026	200.9	Hood
25INR0391	Purple Sage BESS 1	OTH	5/30/2027	156	Collin
25INR0392	Purple Sage BESS 2	OTH	5/30/2027	156	Collin
26INR0543	Three Canes Solar SLF	SOL	12/31/2026	333	Navarro